(1)

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Udemy course
ML, DL, NLP

sec-25
understanding Linear regression

lec-9 Overfitting & Underfitting

lec-10 Linear regression with OLS

(1) Training Dataset
(2) Test Dataset
(3) Validation Dataset

Suppose we have a dataset with 1000 datapoints.
It is divided in 2 parts.

(1) Training Dataset (700)
(2) Test Dataset (300)

We divide our data into training data set & test data set.

Size of House	No. of bedrooms	Price
—	—	—
—	—	—
—	—	—
—	—	—

Here we'll be able to understand overfitting & underfitting.

Q:- Why are we dividing into training & test.

~~80%~~ ~~1000~~

training data

↳ For training the model

Test model → we test model here & model never knows about these specific data points.

Training Dataset in 2 parts

- Train → To train the model
- validation → For Hyper parameter tuning of our model

We combine train & val model at the end to make model performance better.

We have train & test ~~data~~ :- data which works as :-

Train :- Very Good Accuracy

Test :- Very Good Accuracy

⇓

This is a generalised model

~~W.r.t ...~~

But suppose there's another scenario:

Train :- Very Good Accuracy

Test :- Very Bad Accuracy

Since model is giving good accuracy on train data & bad accuracy on test data

⟹ This condition is known as that model is overfitting.

· With new data points, it's not able to predict

Very Good Accuracy [Low Bias]

Bad Accuracy [High Variance]

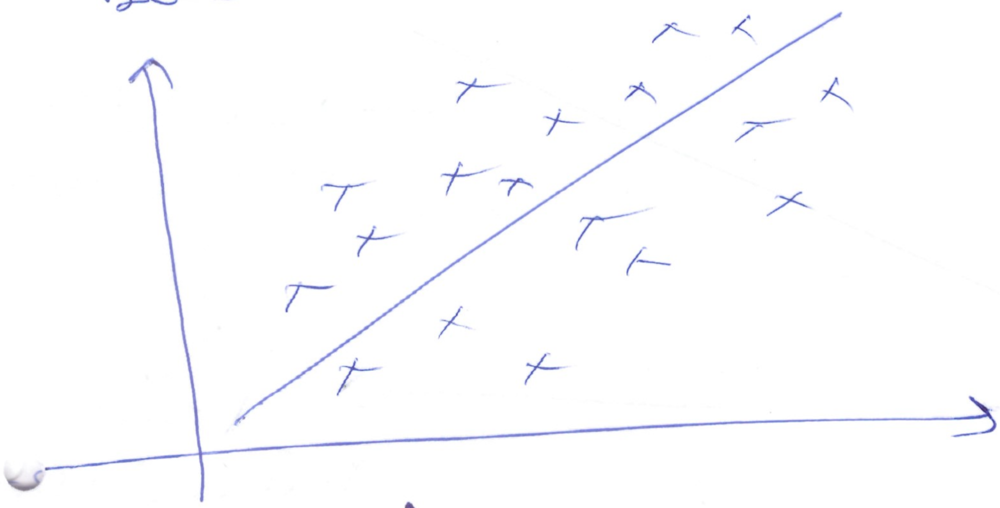
(*) Suppose, there's another scenario where:

Train data :- Model Accuracy is low

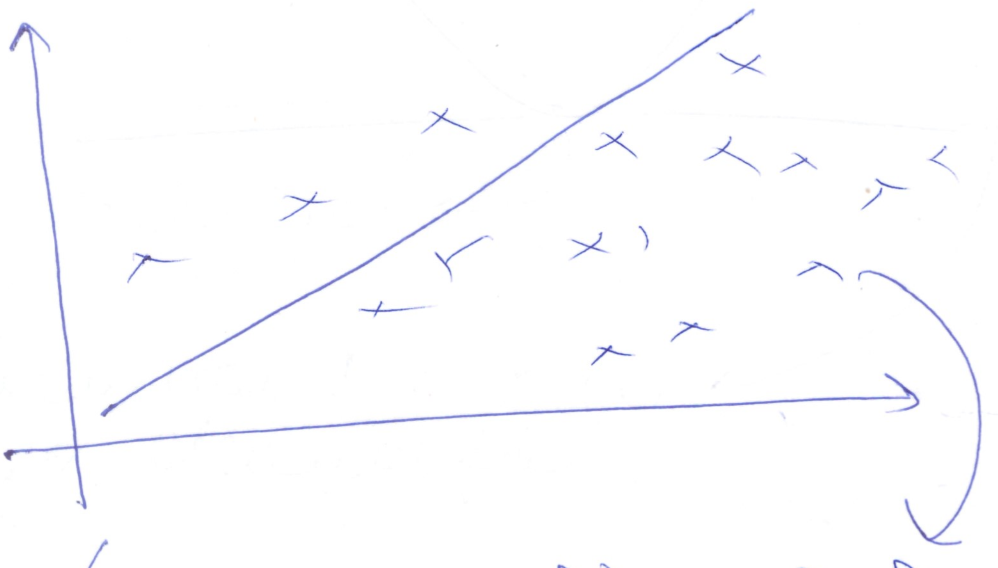
Test data :- Model Accuracy is High

⇓

Model is underfitting



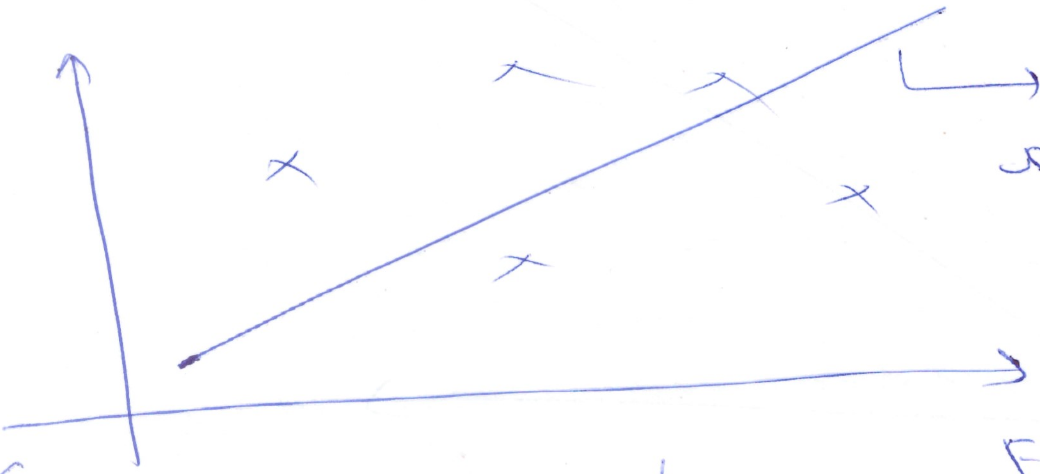
Generalised Model



New data comes

Here its over fitting as the perfect fitting for train data but not test data

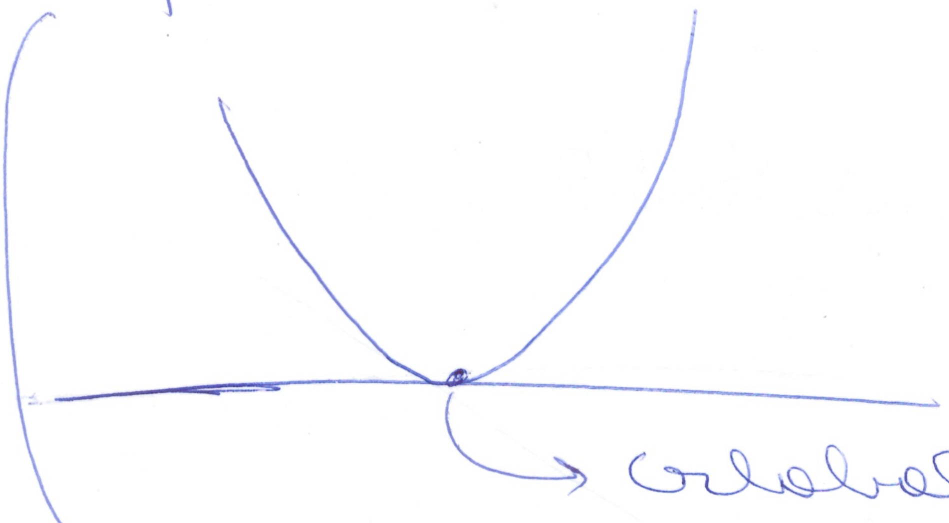
Linear regression using OLS (ordinary least square)



optimization

⇓

Loss Function



Global maximum

OLS ⇒ Formula & calculate B_0 & B_1

OLS → Formula & Calculation B_0 & B_1

7

Ordinary Least Square

$$S(\beta_0, \beta_1) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \beta_0 + \beta_1 x_i\right)^2$$

$$S(\beta_0, \beta_1) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \beta_0 - \beta_1 x_i\right)^2$$

$$\Downarrow$$

$$\frac{\partial S}{\partial \beta_0}(\beta_0, \beta_1) = \frac{2}{n}\sum_{i=1}^{n}\left(y_i - \beta_0 - \beta_1 x_i\right)(0 - 1 - 0)$$

$$= \frac{-2}{n}\sum_{i=1}^{n}\left(y_i - \beta_0 - \beta_1 x_i\right) = 0$$

$$\frac{\partial S}{\partial \beta_1}(\beta_0, \beta_1) = \frac{2}{n}\sum_{i=1}^{n}\left(y_i - \beta_0 - \beta_1 x_i\right)(-x_i)$$

$$= \frac{-2}{n}\sum_{i=1}^{n}\left(y_i - \beta_0 - \beta_1 x_i\right)^{=0}(x_i) = 0$$

$\longrightarrow$ 2

p. 7

$$-\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\Rightarrow -\sum_{i=1}^{n} y_i + n \times \beta_0 + \beta_1 \sum_{i=1}^{n} x_i = 0$$

$$n\beta_0 + \beta_1 \bar{x}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$\frac{\partial S}{\partial \beta_0}(\beta_0, \beta_1) = \frac{-2}{n} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\frac{\partial S}{\partial \beta_1}(\beta_0, \beta_1) = \frac{2}{n} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)(-x_i) = 0$$

$$\boxed{\beta_0 = \bar{y} - \beta_1 \bar{x}}$$

→ Intercept.

equation 2:

$$\frac{-2}{n}\sum_{i=1}^{n}(y_i-\beta_0-\beta_1 x_i)(x_i)=0$$

$$\Downarrow$$

$$\sum_{i=1}^{n}(y_i-\beta_0-\beta_1 x_i)(x_i)=0$$

$$\Downarrow$$

$$\sum_{i=1}^{n}x_i y_i-\beta_0\sum_{i=1}^{n}(x_i)-\beta_1\sum_{i=1}^{n}(x_i)^2=0$$

$$\downarrow$$

Replace $\beta_0=\bar{y}-\beta_1\bar{x}$

$$\sum_{i=1}^{n}(x_i y_i-(\bar{y}-\beta_1\bar{x})x_i-\beta_1 x_i^2)=0$$

$$\sum_{i=1}^{n}\beta_1(\bar{x}-x_i)=-\sum_{i=1}^{n}(y_i-\bar{y})$$

(B)

$$B_1 = \frac{\sum_{i=1}^{n} (y_i - \bar{y})}{\sum_{i=1}^{n} (x_1 - \bar{x})}$$

OLS

$$B_0 = \bar{y} - B_1 \bar{x}$$

~~Ice~~

intercept